

# Control Algorithm Performance Evaluation

AI-POWER

Version 1.1

## Abstract

This document establishes a standardized evaluation framework for comparing control algorithms in power electronic converters. The protocol covers four key performance metrics: Setting Time, Computation Time, Robustness, and Total Harmonic Distortion (THD). Evaluation is conducted on both MATLAB/Simulink simulation platform and dSPACE real-time hardware with 10 kHz switching frequency. All scoring criteria are referenced to industry standards and manufacturer specifications.

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## 1 Introduction

This evaluation protocol provides a systematic approach to assess and compare control algorithms for power electronic applications. The framework ensures fair comparison by standardizing test conditions, measurement procedures, and scoring criteria.

### 1.1 Evaluation Platforms

- **Simulation Platform:** MATLAB R2023b/Simulink with Simscape Electrical
- **Hardware Platform:** dSPACE DS1202
- **Switching Frequency:** 10 kHz (period = 100  $\mu$ s)
- **Sampling Frequency:** 10 kHz

## 1.2 Reference Standards

All evaluation criteria are aligned with:

- IEEE Std 519-2014: Harmonic limits
- TI TMS320F2833x Technical Reference Manual (SPRU430F)
- dSPACE DS1202 Hardware Installation and Configuration Guide
- MathWorks MATLAB/Simulink Best Practices

## 2 General Scoring Methodology

### 2.1 Scoring Scale

All metrics are normalized to a 0-1 scale:

Table 1: Scoring Scale Interpretation

| Score   | Performance Level                 |
|---------|-----------------------------------|
| 1.0     | Excellent (Top tier performance)  |
| 0.8-0.9 | Good (Exceeds requirements)       |
| 0.6-0.7 | Satisfactory (Meets requirements) |
| 0.4-0.5 | Fair (Needs improvement)          |
| 0.0-0.3 | Poor/Unacceptable                 |

## 3 Detailed Scoring Criteria

### 3.1 Settling Time ( $t_s$ )

**Definition:** Time required for the system output to reach and remain within  $\pm 2\%$  of the final steady-state value after a step reference change.

**Test Condition:** Torque step change .

**Scoring Table:**

Table 2: Settling Time Scoring Criteria

| Score | Time Range                               | Switching Cycles | Grade | Justification & Reference                                                    |
|-------|------------------------------------------|------------------|-------|------------------------------------------------------------------------------|
| 1.0   | $t_s < 1 \text{ ms}$                     | 10 cycles        | A+    | TI C2000™ DMCs recommend $\leq 10 \text{ ms}$ for motor control applications |
| 0.9   | $1 \text{ ms} \leq t_s < 2 \text{ ms}$   | 10-20 cycles     | A     | Excellent dynamic response for 10 kHz systems                                |
| 0.8   | $2 \text{ ms} \leq t_s < 4 \text{ ms}$   | 20-40 cycles     | B+    | Good response, meets most industry requirements                              |
| 0.6   | $4 \text{ ms} \leq t_s < 8 \text{ ms}$   | 40-80 cycles     | B     | Acceptable for non-critical applications                                     |
| 0.4   | $8 \text{ ms} \leq t_s < 12 \text{ ms}$  | 80-120 cycles    | C     | Slow response, may affect system performance                                 |
| 0.2   | $12 \text{ ms} \leq t_s < 20 \text{ ms}$ | 120-200 cycles   | D     | Poor response, requires improvement                                          |
| 0.0   | $t_s \geq 20 \text{ ms}$                 | 200 cycles       | F     | Unacceptable for power electronic systems                                    |

Table 3: Overshoot Scoring Criteria for Load Torque Step Change

| Score | Overshoot (%)          | Oscillations  | Grade | Assessment            |
|-------|------------------------|---------------|-------|-----------------------|
| 1.0   | $M_p \leq 5\%$         | Negligible    | A+    | Servo-grade damping   |
| 0.9   | $5\% < M_p \leq 10\%$  | 1-2 minor     | A     | Excellent, industrial |
| 0.8   | $10\% < M_p \leq 15\%$ | 2-3 decaying  | B+    | Good performance      |
| 0.7   | $15\% < M_p \leq 20\%$ | 3-4 visible   | B     | Acceptable            |
| 0.5   | $20\% < M_p \leq 30\%$ | 4-5 sustained | C     | Needs tuning          |
| 0.3   | $30\% < M_p \leq 40\%$ | 5 significant | D     | High risk             |
| 0.0   | $M_p > 40\%$           | Divergent     | F     | Unstable              |

### 3.2 Computation Speed ( $t_c$ )

**Definition:** Worst-case execution time (WCET) for one control cycle implementation.

**Measurement Method:** Using dSPACE Trace functionality and CPU timer interrupts.

**Scoring Table:**

Table 4: Computation Time Scoring Criteria

| Score | Time Range                                 | CPU Utilization | Grade | Justification & Reference                |
|-------|--------------------------------------------|-----------------|-------|------------------------------------------|
| 1.0   | $t_c < 20 \mu\text{s}$                     | $< 20\%$        | A+    | Leaves ample margin for other tasks      |
| 0.9   | $20 \mu\text{s} \leq t_c < 30 \mu\text{s}$ | 20-30%          | A     | Excellent real-time capability           |
| 0.8   | $30 \mu\text{s} \leq t_c < 40 \mu\text{s}$ | 30-40%          | B+    | Good, typical for predictive controllers |
| 0.7   | $40 \mu\text{s} \leq t_c < 50 \mu\text{s}$ | 40-50%          | B     | Acceptable for 10 kHz switching          |
| 0.6   | $50 \mu\text{s} \leq t_c < 60 \mu\text{s}$ | 50-60%          | C+    | Moderate margin available                |
| 0.5   | $60 \mu\text{s} \leq t_c < 70 \mu\text{s}$ | 60-70%          | C     | Limited margin, requires optimization    |
| 0.4   | $70 \mu\text{s} \leq t_c < 80 \mu\text{s}$ | 70-80%          | D+    | Critical timing, difficult to maintain   |
| 0.3   | $80 \mu\text{s} \leq t_c < 90 \mu\text{s}$ | 80-90%          | D     | Very tight timing constraints            |
| 0.2   | $90 \mu\text{s} \leq t_c < 95 \mu\text{s}$ | 90-95%          | F+    | Risky, may cause missed deadlines        |
| 0.0   | $t_c \geq 95 \mu\text{s}$                  | 95%             | F     | Unreliable for real-time control         |

#### Platform Specifications:

- dSPACE DS1202: PPC 750GX @ 1 GHz
- Maximum allowable:  $0.9 \times T_s = 90 \mu\text{s}$  for safety margin

#### Reference:

- dSPACE Real-Time Interface (RTI) Implementation Guide

### 3.3 Robustness ( $R$ )

**Definition:** System's ability to maintain stability and acceptable performance under parameter variations and model mismatches.

#### Evaluation Components:

1. **Parameter Variation (60% weight):** L, C, R variations from nominal

## 2. Model Mismatch (40% weight): Unmodeled dynamics, linearization errors

### Test Protocol:

- Parameter sweep:  $\pm 50\%$  from nominal values
- Model mismatch: 20% reduction in system order
- Performance measured by IAE (Integral of Absolute Error)

### Scoring Table:

Table 5: Robustness Scoring Criteria

| Score | Stability              | Performance Loss | Grade | Justification & Reference                  |
|-------|------------------------|------------------|-------|--------------------------------------------|
| 1.0   | Always stable          | $< 10\%$         | A+    | Excellent robustness, industrial grade     |
| 0.9   | Always stable          | 10-15%           | A     | Very good, suitable for harsh environments |
| 0.8   | Always stable          | 15-25%           | B+    | Good, meets automotive standards           |
| 0.7   | Stable with margin     | 25-35%           | B     | Acceptable for most applications           |
| 0.6   | Stable with margin     | 35-45%           | C+    | Moderate robustness                        |
| 0.5   | Marginally stable      | 45-60%           | C     | Limited robustness, needs care             |
| 0.4   | Occasional instability | 60-80%           | D+    | Poor, requires careful tuning              |
| 0.3   | Frequently unstable    | 80-100%          | D     | Very poor, not recommended                 |
| 0.2   | Unstable in 50% cases  | 100%             | F+    | Unacceptable performance                   |
| 0.0   | Always unstable        | System failure   | F     | Complete lack of robustness                |

### Scoring Formula:

$$R = 0.6 \times \text{Param\_Score} + 0.4 \times \text{Model\_Score}$$

$$\text{Param\_Score} = \frac{1}{n} \sum_{i=1}^n \left( 1 - \frac{\text{IAE}_i - \text{IAE}_{\text{nominal}}}{\text{IAE}_{\text{nominal}}} \right)$$

### Reference:

- TI Application Report SPRAC11: "Robust Control of Power Converters"

## 3.4 Total Harmonic Distortion (THD)

### Definition:

$$\text{THD} = \frac{\sqrt{\sum_{h=2}^{50} V_h^2}}{V_1} \times 100\%$$

where  $V_h$  is the RMS voltage of harmonic  $h$ , and  $V_1$  is the fundamental voltage.

### Measurement:

- FFT analysis with 4096 points
- Hanning window, frequency resolution: 12.5 Hz
- Steady-state conditions, rated load

**Scoring Table:**

Table 6: THD Scoring Criteria (IEEE 519-2014 Based)

| Score | THD Range |           | IEEE Class           | Grade | Justification & Reference             |
|-------|-----------|-----------|----------------------|-------|---------------------------------------|
| 1.0   | THD < 3%  |           | Special Applications | A+    | Exceeds medical/military standards    |
| 0.9   | 3%        | THD < 5%  | Class 1              | A     | Excellent power quality               |
| 0.8   | 5%        | THD < 8%  | Class 2              | B+    | Good, meets hospital standards        |
| 0.7   | 8%        | THD < 10% | Class 3              | B     | Acceptable for general systems        |
| 0.6   | 10%       | THD < 12% | Class 4              | C+    | Satisfactory, meets industrial limits |
| 0.5   | 12%       | THD < 15% | Class 5              | C     | Fair, borderline acceptable           |
| 0.4   | 15%       | THD < 20% | -                    | D+    | Poor, may affect sensitive equipment  |
| 0.3   | 20%       | THD < 25% | -                    | D     | Very poor, requires filtering         |
| 0.2   | 25%       | THD < 30% | -                    | F+    | Unacceptable for most applications    |
| 0.0   | THD ≥ 30% |           | -                    | F     | Violates power quality standards      |

**Reference:**

- IEEE Std 519-2014: "IEEE Recommended Practice and Requirements for Harmonic Control"
- IEC 61000-3-2: Limits for harmonic current emissions
- TI Application Report SLAA602: "Power Quality Measurements"